

Modular Programming 2-oyligi

2 - oy uchun plan

- 1) 4 ta Project (data extraction + Deployment)
- 2) Modelni rivojlantirish => (tekshirish, neanelar)
- 3) Modular Programmingni ham rivojlantirish (structure)
- 4) Git bash commands
- 5) Multi-class classification
- 6) Multi-output models
- 7) Deployment

Asosiy qobiliyatlarni shakilllantirish + rivojlantirish:

- Problem solving (codewars or leetcode) ^{at least one problem each day}
- To'g'ri va tez yozish (monkextype)
- Logical thinking (kitob o'qish) (olinda borli)

Multi-class classification

- Binary Classification => bu 2 ta classificationdan iborat bo'ladi
- ↳ (Yes or No, 0 or 1, false or True)

Binary Classification

No.

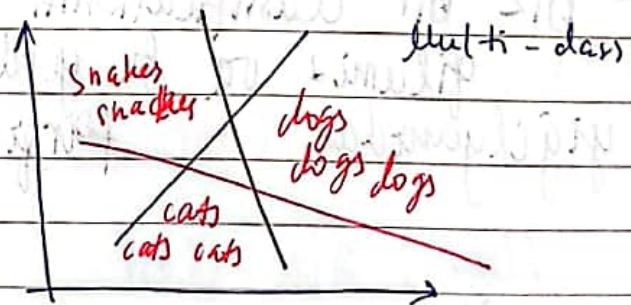
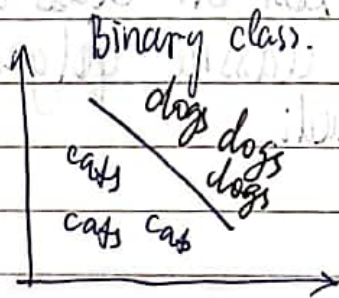
- 2 ta class li target o'qimat mavjud bo'lgan dataset k.k bo'ladi.
- Confusion matrixni tushunish oson (0/1)
- 0/1 ligi uchun modelni tushunish oson.
- Faqat bitta classifier modelni talab qiladi.
- One vs another (0/1, yes/no)

OVA

Multi-class classification

- Logistic Regression (o'qimda sigmoid function or
logit function (ova oraliqida) → Binary Classification
(DT, RF → Binary and Multi class)

- Xaqiqiy hayotda ko'plab xolatlar multiclass



- data kutta va balansli bo'lish k.k

Yechim

- Multi-classlarni ko'plab binaryga xolatga bo'lib olish.

- 2 ta asosiy strategiya mavjud:

— One-vs-Rest (OvR)

0

1
it vs (ilon + mushuk)

0
mushuk vs (it + ilon)

0
ilon vs (it + mushuk)

— One-vs-One (OvO) usulida har bir klass boshqa har bir klass bilan alohida model orqali solishtiriladi.

1. cat vs dog

2. cat vs snake

3. snake vs dog

$$= \frac{k(k-1)}{2}$$

- OvR

— Biz bir classification modelni xar bir class u-n train qilamiz va bu yerda u bir classni qolgan/ini yig'ilganlardan (rest) farqini o'rganadi

misol: 3 ta class

— ilon

— It

— mushuk

• Model 1: ilon vs (It + mushuk)

• Model 2: It vs (ilon + mushuk)

• Model 3: mushuk vs (It + ilon)

Agar classimiz N ta bo'lsa N marta train qilamiz.

Bu yerda biz 3ta modelni train qilamiz (1ta algorithm)

OvR

↳ Prediction

↳ Model 1 : $\text{dlon} \text{ vs } (\text{It} + \text{llushuk}) \rightarrow 0.65$

↳ Model 2 : $\text{It} \text{ vs } (\text{dlon} + \text{llushuk}) \rightarrow 0.18$

↳ Model 3 : $\text{llushuk} \text{ vs } (\text{It} + \text{dlon}) \rightarrow 0.25$

↓
Predicted class : dlon

OvR : Example 1

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.multiclass import OneVsRestClassifier,
OneVsOneClassifier
from sklearn.metrics import classification_report
```

```
X, y = load_iris(return_X_y=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```

⋮


```
ovr-model = OneVsRestClassifier(LogisticRegression(
    max_iter=1000, random_state=42))
```

```
ovr-model.fit(X_train, y_train)
```

```
# Predict and evaluate
```

```
y_pred_ovr = ovr_model.predict(X_test)
print("=== One-vs-Rest (OvR) ===")
print(classification_report(y_test, y_pred_ovr))
```

OvR + (advantages)

- ↳ Sodda va oson ishlatish
- ↳ Agar classlar ozroq bo'lsa tez ishlaydi.

OvR - (disadvantages)

- ↳ Imbalanced dataset/ b/n yaxshi ishlamaydi.

OvO : One-vs-One Classifier

Bu yerda biz classification modellarni har bir class uchun **train qilamiz** va bu yerda har bir class'ni **golgantari bilan yakkama yakkaga farqini o'rganamiz**.

misol : 3 ta class

- I lon
- It
- blushuk

Model 1: I lon vs it

Model 2: I lon vs blushuk

Model 3: It vs blushuk

OvO: Example 2

```
ovo_model = OneVsOneClassifier(LogisticRegression(
    max_iter = 1000, random_state = 42))
```

```
ovo_model.fit(X_train, y_train)
```

```
y_pred_ovo = ovo_model.predict(X_test)
```

```
print("=== One-vs-One (OvO) ===")
```

```
print(classification_report(y_test, y_pred_ovo))
```

OvO formulasi: $\frac{n(n-1)}{2}$

OvO vs OvR farqli asosiy farqi

Modellar sonida va sekin ishlashida

→ OvO :

↳ Prediction

↳ Model 1: Elon vs it (Elon)

↳ Model 2: Elon vs mushuk (mushuk)

↳ Model 3: it vs mushuk (mushuk)



- Predicted class: mushuk
- Agar teng bo'lib qolsa confidence scorega ko'ra eng yuqori score d'gani predict qilinadi.
- Confidence $\Rightarrow$ Ehtimollar nazariyasi

confidence score formulari

→
$$\text{Con. Score} = \frac{1}{1 + e^{-z}} = (z \approx 2.71)$$

$$= \frac{e^z}{e^z + 1} \Rightarrow (0, 1) \text{ oralig'ida bo'ladi.}$$

OvO + (advantages)

- ↳ Imbalanced dataset/da ishlatish yaxshi
- ↳ Hecim classlarda xam yaxshi

OvO - (disadvantages)

- 10 ta classes $\rightarrow$ 45 models $\frac{n(n-1)}{2} = \frac{10(10-1)}{2} = 45$
- 100 ta classes $\rightarrow$ 4950 ta models
- Nimaga olib keladi? $\Rightarrow$ 1. Ortig'cha chiqim, sarfgor olib keladi. 2. Dataset ko'p bo'lsa ko'p vaqt ketadi. 3. Modelni sekin ishlatishga ham olib keladi.

Bu shunchaki biz train qilayotganimizda, lekin hali model tayyorlab olgandan keyin, uni xavfli ishga tushirayotgan paytimizda ham buni monitoring qilish, buni natijasini olishimiz ozinga qiyinchilik tug'diradi.

lekin baribir aytib o'tilganidek bizni MLda algoritim ko'ramizmi biron bir usul, method ko'ramizmi, qaysi birida qiymat yozishi chiqishi u-n albatta ko'proq vaqt talab qiladi.

~~chunki qozon~~

Multi-class classification umumlashtirgan holda

- Ko'plab class outputga ega bo'lgan dataset k-k bol.
- Multi-class model:

- One-vs-Rest (OvR)

- Har bir model bitta klassni boshqalidan ajratib olishga harakat qiladi (yes or no).

- Example: Nafaqa class bo'lsa shuncha modellarni train qilamiz.

- One-vs-One (OvO)

- Har bir ~~class~~ model bitta klassni keyingi class bilan taqqoslab o'rganadi.

- Example: $n(n-1)/2 \Rightarrow 4(4-1)/2 = 6$ ta model train qilindi.

Max - iterations (parameter)

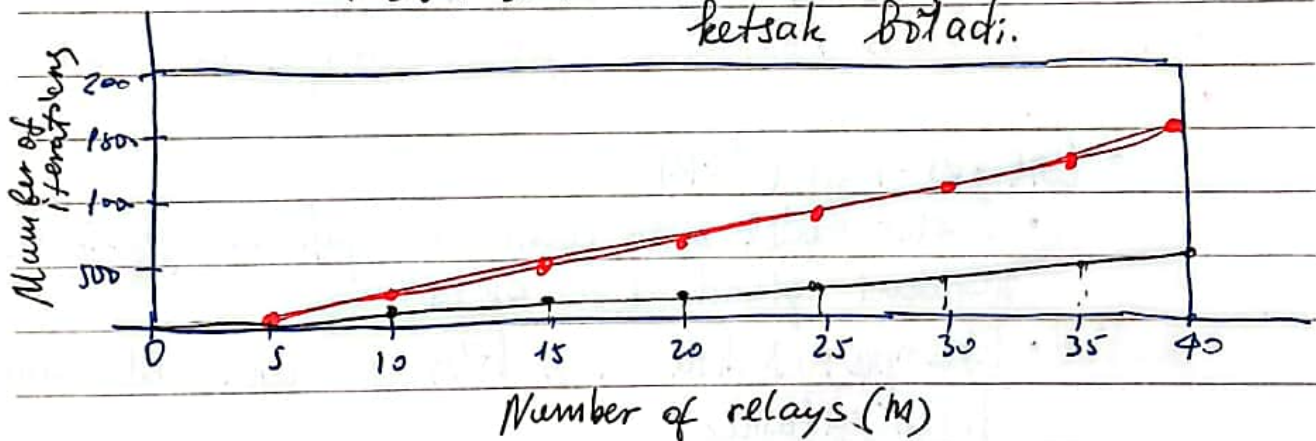
- Manziliimizga ~~h-h-h-h~~ ~~h-h~~ gadam ba - gadam borish uchun gadam $\rightarrow$ Best solution
- Bazida oz bazida kōp gadam h-h bo'lad!

- Barida oz barida kōp gadam k-k bo'lad!

- Algorithms :

- Linear (ML da)
- DL (Deep Learning da)
- Clustering (Unsupervised ML da)

↳ 3ta xolatda ma'it parametrni ishlatib ketsak bo'ladi.



Eng yaxeshi tushuncha:

- Iteration = model = weightlarni yaxshiroq tomonga bir marta yangilashi.
 - Optimization = eng yaxshi weightlarni topish jarayoni.
- max_iter = shu jarayonda modelga berilgan maksimal urinishlar soni.

Logistic Regression (max_iter = 1000)

Logistic Regression (max_iter = 1000)
 "Model, eng yaxshi weight/ni topish u-n 1000 marta o'qish
 univishig mumkin"